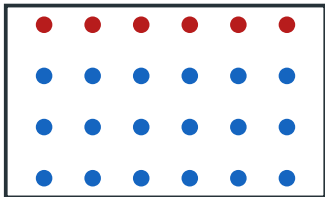


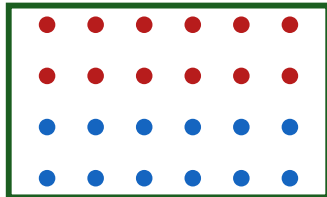
unequal: $n_D = 3n_T$



$$n_D n_T \propto 18 \times 6 = 108$$

● deuteron ● triton

equal: $n_D = n_T$



$$n_D n_T \propto 12 \times 12 = 144$$

Same 24 ions in both boxes. The reaction rate goes as the number of D--T PAIRS, $n_D n_T$, and with $n_D + n_T$ fixed that product is a downward parabola: complete the square,

$$n_D n_T = \frac{1}{4}(n_D + n_T)^2 - \frac{1}{4}(n_D - n_T)^2, \text{ which is largest exactly when } n_D = n_T.$$

Explanatory sketch; schematic, not to scale.